1 For Guest Users

1.1 Utility Tree for Guest User Browsing Posts

Table 1: Utility Tree for Guest User Browsing Posts

Quality Attribute	**Attribute Refinement**	**ASR**
Performance	Page load time	A guest user browses the list of posts under normal load, and the page loads in less than 2 seconds. (H,M)
	Scrolling / pagination	A guest user navigates between pages or scrolls through posts under peak load, and additional content loads in less than 1 second. (H,M)
Usability	Ease of navigation	A first-time guest user browses posts without prior instruction and can easily navigate between posts and lists. (H,M)
	Readability	A guest user views posts on different devices, and the content is displayed clearly with a consistent layout. (M,M)
Availability	Read access	A guest user attempts to browse posts at any time, and the system remains available with 99.9% uptime. (H,H)
Scalability	Traffic handling	At peak load, the system supports a large number of guest users browsing posts simultaneously without noticeable performance degradation. (H,M)
Security	Access restriction	A guest user attempts a restricted action such as posting or commenting, and the system prevents the action and prompts the user to sign up. (H,H)
Reliability	Content consistency	A guest user browses posts during operation, and all content is displayed correctly without missing or corrupted data. (H,M)

1.2 Utility Tree for Guest User Viewing Posts and Comments

Table 2: Utility Tree for Guest User Viewing Posts and Comments

Quality Attribute	**Attribute Refinement**	**ASR**
Performance	Post and comment load time	A guest user opens a post with comments under normal load, and all content is displayed in less than 2 seconds. (H,M)
	Comment expansion	A guest user expands comment threads under peak load, and additional comments load in less than 1 second. (H,M)
Usability	Readability	A guest user reads posts and nested comments, and the content is clearly structured and easy to follow. (H,M)
	Navigation	A guest user navigates between posts and comment threads without confusion or additional guidance. (H,M)
Availability	Content access	A guest user attempts to view posts and comments at any time, and the system remains available with 99.9% uptime. (H,H)
Scalability	Concurrent viewing	At peak load, many guest users view posts and comments simultaneously, and the system maintains stable response time. (H,M)
Security	Access restriction	A guest user attempts to interact (e.g., reply or vote), and the system prevents the action and prompts sign-up. (H,H)
Reliability	Data consistency	A guest user views posts and comments, and all content is displayed correctly without missing or duplicated entries. (H,M)

1.3 Utility Tree for Guest User Searching Keywords

Table 3: Utility Tree for Guest User Searching Keywords

Quality Attribute	**Attribute Refinement**	**ASR**
Performance	Search response time	A guest user enters a keyword under normal load, and the system returns relevant search results in less than 1 second. (H,H)
	Result pagination	A guest user navigates through multiple pages of search results under peak load, and each page loads in less than 1 second. (H,M)
Usability	Search accuracy	A guest user searches using common keywords, and the system returns relevant and meaningful results. (H,H)
	Ease of use	A guest user performs a search without prior instruction and can easily refine or modify the query. (H,M)
Availability	Search availability	A guest user performs a search at any time, and the search functionality is available with 99.9% uptime. (H,H)
Scalability	Query handling	At peak load, many guest users perform searches simultaneously, and the system maintains fast response times without degradation. (H,M)
Security	Query validation	A guest user submits search input, and the system sanitizes input to prevent injection or malicious queries. (H,H)
Reliability	Result consistency	A guest user performs the same search multiple times under stable data conditions, and the system returns consistent results. (M,M)

1.4 Utility Tree for Guest User Searching Communities

Table 4: Utility Tree for Guest User Searching Communities

Quality Attribute	**Attribute Refinement**	**ASR**
Performance	Search response time	A guest user searches for communities under normal load, and the system returns results in less than 1 second. (H,H)
	Result loading	A guest user navigates through community search results under peak load, and additional results load in less than 1 second. (H,M)
Usability	Discoverability	A guest user searches for communities using keywords, and the system returns relevant communities based on interests. (H,H)
	Ease of navigation	A guest user can easily browse, filter, and select communities from search results without prior instruction. (H,M)
Availability	Search access	A guest user performs community search at any time, and the system remains available with 99.9% uptime. (H,H)
Scalability	Concurrent search	At peak load, many guest users search for communities simultaneously, and the system maintains stable response time. (H,M)
Security	Input validation	A guest user submits search queries, and the system validates and sanitizes input to prevent malicious usage. (H,H)
Reliability	Result consistency	A guest user performs repeated searches under stable conditions, and the system returns consistent community results. (M,M)

1.5 Utility Tree for Guest User Viewing User Profiles

Table 5: Utility Tree for Guest User Viewing User Profiles

Quality Attribute	**Attribute Refinement**	**ASR**
Performance	Profile load time	A guest user opens a user profile under normal load, and the profile page loads in less than 2 seconds. (H,M)
	Content loading	A guest user views posts or activity on a profile under peak load, and content loads in less than 1 second. (H,M)
Usability	Information clarity	A guest user views a profile and can easily understand the user's bio, posts, and activity. (H,M)
	Navigation	A guest user navigates between profile sections (posts, comments, etc.) without confusion. (H,M)
Availability	Profile access	A guest user attempts to view profiles at any time, and the system remains available with 99.9% uptime. (H,H)
Scalability	Concurrent access	At peak load, many guest users view profiles simultaneously, and the system maintains stable response time. (H,M)
Security	Privacy protection	A guest user views a profile, and the system ensures only public information is displayed while restricted data is protected. (H,H)
Reliability	Data consistency	A guest user views a profile, and all displayed information is accurate and not missing or corrupted. (H,M)

1.6 Utility Tree for Guest User Sign-up Prompt on Restricted Actions

Table 6: Utility Tree for Guest User Sign-up Prompt on Restricted Actions

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Clarity of prompt	A guest user attempts a restricted action, and the system displays a clear and understandable sign-up prompt explaining the restriction. (H,H)
	User guidance	A guest user sees the prompt and can easily proceed to the sign-up process without confusion. (H,M)
Performance	Prompt response time	A guest user triggers a restricted action under normal load, and the sign-up prompt appears in less than 0.5 seconds. (H,M)
Availability	Prompt availability	A guest user attempts restricted actions at any time, and the sign-up prompt mechanism is always available. (H,H)
Security	Access enforcement	A guest user attempts restricted actions, and the system consistently blocks the action and enforces authentication requirements. (H,H)
Reliability	Prompt consistency	A guest user repeatedly attempts restricted actions, and the system consistently displays the correct prompt without failure. (M,M)

1.7 Utility Tree for Guest User Sign-up with Email and OTP Verification

Table 7: Utility Tree for Guest User Sign-up with Email and OTP Verification

Quality Attribute	**Attribute Refinement**	**ASR**
Security	OTP verification	A guest user signs up with email, and the system generates and sends a one-time password (OTP) that must be correctly validated before account creation. (H,H)
	Data protection	A guest user submits personal information during sign-up, and the system securely stores the data using encryption. (H,H)
Performance	OTP delivery time	A guest user requests an OTP under normal load, and the system delivers the OTP within 5 seconds. (H,M)
	Registration response	A guest user completes OTP verification, and the account is created in less than 2 seconds. (H,M)
Usability	Ease of registration	A guest user completes the sign-up and OTP verification process without prior instruction and minimal steps. (H,M)
Availability	Registration access	A guest user attempts to sign up at any time, and the registration and OTP services are available with 99.9% uptime. (H,H)
Reliability	OTP correctness	A guest user enters a valid OTP within the allowed time, and the system consistently verifies it correctly without errors. (H,M)
Scalability	Concurrent registrations	At peak load, many guest users register simultaneously, and the system maintains stable performance for OTP generation and verification. (M,M)

2 For Members

2.1 Utility Tree for Member Sign-in with External Authentication Provider

Table 8: Utility Tree for Member Sign-in with External Authentication Provider

Quality Attribute	**Attribute Refinement**	**ASR**
Security	Third-party authentication	A member chooses to sign in using an external authentication provider, and the system securely delegates authentication without exposing user credentials. (H,H)
	Token handling	A member successfully authenticates via the external provider, and the system receives and securely processes an authentication token containing user identity information. (H,H)
	Token validation	A member sends requests after authentication, and the system validates the token's integrity and expiration before granting access. (H,H)
Performance	Login response time	A member signs in using an external provider under normal load, and access is granted within 2 seconds. (H,M)
	Redirection latency	A member is redirected between the platform and the external provider, and the transitions occur without noticeable delay. (M,M)
Usability	Ease of login	A member signs in using an external provider with minimal steps and without needing to create or remember a password. (H,H)
Availability	External dependency	A member attempts to sign in, and the system handles the availability of the external authentication service gracefully without blocking system access. (H,H)
Reliability	Authentication consistency	A member repeatedly signs in using the external provider under stable conditions, and the system consistently authenticates without failure. (H,M)
Interoperability	External integration	The system communicates with external authentication services using standard protocols to exchange authentication data correctly. (H,H)
Manageability	Monitoring login service	Admins can monitor login success/failure rates via dashboards. (M,M)
	Error diagnostics	When login fails, logs provide sufficient detail for debugging authentication issues. (M,M)
Flexibility	Multiple login methods	The system supports adding new authentication providers without major redesign. (M,M)

2.2 Utility Tree for Member Sign-in with Email and Password

Table 9: Utility Tree for Member Sign-in with Email and Password

Quality Attribute	**Attribute Refinement**	**ASR**
Security	Credential protection	A member enters email and password, and the system securely transmits credentials over a protected channel and stores passwords using strong hashing mechanisms. (H,H)
	Token-based authentication	A member successfully authenticates, and the system generates a secure authentication token containing user identity and authorization data. (H,H)
	Token validation	A member sends requests with an authentication token, and the system validates the token's integrity and expiration before granting access. (H,H)
Performance	Login response time	A member signs in under normal load, and access is granted within 2 seconds. (H,M)
	Authentication overhead	A member makes authenticated requests, and token validation adds minimal overhead to each request. (M,M)
Usability	Session continuity	A member remains authenticated across multiple interactions without needing to repeatedly enter credentials. (H,M)
Availability	Authentication service	A member attempts to sign in at any time, and the authentication service is available with 99.9% uptime. (H,H)
Reliability	Authentication consistency	A member repeatedly signs in with correct credentials, and the system consistently authenticates without failure. (H,M)
Scalability	Concurrent authentication	At peak load, many members authenticate and send requests simultaneously, and the system maintains stable performance without degradation. (H,M)

2.3 Utility Tree for Member Profile Customization

Table 10: Utility Tree for Profile Customization

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Profile editing ease	A member updates bio, avatar, and banner, and the system allows completion without guidance in a few intuitive steps. (H,M)
	Feedback clarity	A member performs profile updates, and the system provides clear success or error messages. (H,M)
Performance	Upload response time	A member uploads avatar or banner images under normal load, and the system processes and displays them within 3 seconds. (H,M)
	Profile load time	A member views a profile with images, and the page loads within 2 seconds. (H,M)
Security	File validation	A member uploads an image, and the system validates file type and size to prevent malicious content. (H,H)
	Data protection	A member updates profile information, and the system ensures only the owner can modify their profile data. (H,H)
Reliability	Data consistency	A member updates profile data, and all changes are saved correctly without data loss or corruption. (H,M)
Availability	Profile editing access	A member attempts to edit their profile at any time, and the system remains available with 99.9% uptime. (H,H)
Scalability	Concurrent uploads	At peak load, many members upload avatars and banners simultaneously, and the system maintains stable performance. (M,M)
Maintainability	Profile module modification	Developers modify profile customization features, and changes can be made without affecting unrelated modules. (M,M)
Flexibility	Profile extensibility	New profile fields (e.g., social links) can be added without significant redesign. (M,M)
Supportability	Error logging	Failures during profile updates or uploads are logged to support debugging and issue resolution. (M,M)
Testability	Feature testing	Profile customization features can be verified through automated and manual test cases. (M,M)

2.4 Utility Tree for Post Management

Table 11: Utility Tree for Member Post Management

Quality Attribute	Attribute Refinement	ASR
Usability	Content interaction ease	A member creates, edits, or deletes a post (text, image, poll), and the system allows completion in a few intuitive steps with clear feedback. (H,M)
	Input validation feedback	A member submits or updates a post with missing or invalid data, and the system provides clear and immediate feedback. (H,M)
Performance	Post submission time	A member submits a text post under normal load, and the system publishes it within 1 second. (H,M)
	Image upload processing	A member uploads images to a post, and the system processes and attaches them within 3 seconds. (H,M)
	Poll creation response	A member creates or updates a poll, and the system registers poll options and updates the post within 1 second. (H,M)
	Edit/Delete response time	A member edits or deletes a post, and the system processes the action within 1 second. (H,M)
Security	Content validation	A member submits or updates post content, and the system sanitizes inputs to prevent malicious scripts or harmful data. (H,H)
	Access control	A member attempts to modify or delete a post, and only the owner or authorized roles are allowed to perform the action. (H,H)
Reliability	Data persistence	A member creates, edits, or deletes a post, and the system ensures all changes are stored correctly without data loss or inconsistency. (H,H)
	Poll result correctness	Members vote on a poll, and the system accurately records votes and consistently displays correct aggregated results. (H,H)
Availability	Posting service availability	A member attempts to create, edit, or delete a post at any time, and the posting service is available with 99.9% uptime. (H,H)
Scalability	Concurrent post operations	At peak load, many members create, edit, delete posts, and vote on polls simultaneously, and the system maintains stable performance. (H,M)
Maintainability	Post feature evolution	Developers extend or modify post-related features without affecting existing functionality. (M,M)
Flexibility	Content type extensibility	The system supports adding new post types or poll features without major redesign. (M,M)
Supportability	Error logging	Failures during post operations or poll interactions are logged for debugging and issue resolution. (M,M)

2.5 Utility Tree for Post Drafting

Table 12: Utility Tree for Saving Post as Draft

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Draft saving ease	A member saves a post as a draft, and the system allows the action with a single step and clear feedback. (H,M)
	Draft retrieval	A member accesses saved drafts, and the system displays and allows editing without confusion. (H,M)
Performance	Save response time	A member saves a draft under normal load, and the system stores it within 1 second. (H,M)
	Load draft time	A member opens a saved draft, and the system loads the content within 1 second. (H,M)
Security	Access control	A member attempts to view or edit a draft, and only the owner can access and modify it. (H,H)
Reliability	Draft persistence	A member saves a draft, and the system ensures the content is stored correctly without data loss. (H,H)
Availability	Draft service availability	A member attempts to save or access drafts at any time, and the system is available with 99.9% uptime. (H,H)
Scalability	Concurrent draft operations	At peak load, many members save and edit drafts simultaneously, and the system maintains stable performance. (M,M)

2.6 Utility Tree for Member Voting on Posts and Comments

Table 13: Utility Tree for Voting (Upvote/Downvote)

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Interaction simplicity	A member upvotes or downvotes a post or comment, and the system allows the action with a single click and immediate visual feedback. (H,M)
	Feedback clarity	A member performs a vote action, and the system clearly reflects the updated vote state and count. (H,M)
Performance	Vote response time	A member votes on a post or comment under normal load, and the system updates the vote within 0.5 seconds. (H,M)
	Count update latency	A member performs voting under peak load, and vote counts are updated and displayed within 1 second. (H,M)
Security	Access control	A member attempts to vote, and only authenticated users are allowed to perform voting actions. (H,H)
	Vote integrity	A member attempts to vote multiple times on the same item, and the system enforces one vote per user per item. (H,H)
Reliability	Vote consistency	A member votes on a post or comment, and the system consistently records and reflects the correct vote state. (H,H)
Availability	Voting service availability	A member attempts to vote at any time, and the voting functionality is available with 99.9% uptime. (H,H)
Scalability	High-frequency interactions	At peak load, many members perform voting simultaneously, and the system maintains stable performance without degradation. (H,M)

2.7 Utility Tree for Commenting and Replying

Table 14: Utility Tree for Comment Creation, Modification, Deletion, and Reply

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Comment interaction ease	A member creates, edits, deletes, or replies to a comment, and the system allows the action in a few intuitive steps with clear feedback. (H,M)
	Threaded conversation clarity	A member replies to a specific comment, and the system displays the conversation in a clear hierarchical structure. (H,M)
Performance	Comment submission time	A member submits or edits a comment under normal load, and the system processes the action within 1 second. (H,M)
	Thread loading time	A member views a post with nested comments, and the system loads and renders the comment thread within 2 seconds. (H,M)
Security	Access control	A member attempts to modify or delete a comment, and only the owner or authorized roles can perform the action. (H,H)
	Content validation	A member submits a comment, and the system sanitizes input to prevent malicious or harmful content. (H,H)
Reliability	Comment consistency	A member performs comment operations, and the system ensures all changes are saved correctly without duplication or loss. (H,H)
Availability	Comment service availability	A member attempts to interact with comments at any time, and the system is available with 99.9% uptime. (H,H)
Scalability	High interaction load	At peak load, many members create and reply to comments simultaneously, and the system maintains stable performance. (H,M)

2.8 Utility Tree for Follow

Table 15: Utility Tree for Following Users and Viewing Their Post History

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Follow interaction ease	A member follows or unfollows another user, and the system completes the action with a single step and clear feedback. (H,M)
	History readability	A member views a followed user's post history, and the system presents posts in a clear and chronological order. (H,M)
Performance	Follow action response	A member follows or unfollows a user under normal load, and the system processes the action within 1 second. (H,M)
	History load time	A member opens a followed user's post history, and the system loads posts within 2 seconds. (H,H)
Security	Access control	A member attempts to follow a user or view their posts, and the system enforces authentication and respects visibility settings. (H,H)
	Data privacy	A member views a user's post history, and the system ensures only accessible posts are displayed. (H,H)
Reliability	Follow consistency	A member follows or unfollows a user, and the system consistently updates and reflects the relationship. (H,H)
	History correctness	A member views post history, and the system consistently displays all posts without duplication or omission. (H,H)
Availability	History access availability	A member attempts to view a user's post history at any time, and the system remains available with 99.9% uptime. (H,H)
Scalability	Concurrent access	At peak load, many members follow users and view post histories simultaneously, and the system maintains stable performance. (H,M)
Maintainability	Feature updates	Developers modify follow or history viewing features without affecting other system components. (M,M)
Flexibility	History extensibility	The system supports future enhancements such as filtering or sorting post history without major redesign. (M,M)
Supportability	Activity logging	Follow actions and history access operations are logged to support debugging and monitoring. (M,M)

2.9 Utility Tree for Member Notification System

Table 16: Utility Tree for Notifications

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Notification clarity	A member receives a notification, and the system displays clear and understandable information about the triggering event. (H,M)
	Interaction ease	A member views a notification and can easily navigate to the related content with a single action. (H,M)
Performance	Delivery latency	An event that triggers a notification occurs under normal load, and the system delivers the notification within 1 second. (H,H)
	Retrieval time	A member opens the notification list, and the system loads notifications within 1 second. (H,M)
Reliability	Delivery guarantee	A notification event occurs, and the system ensures the notification is delivered without loss. (H,H)
	Consistency	A member views notifications, and the system consistently displays correct and up-to-date notification states. (H,H)
Availability	Notification service availability	A member receives notifications at any time, and the notification system is available with 99.9% uptime. (H,H)
Scalability	High event throughput	At peak load, many events trigger notifications simultaneously, and the system maintains stable performance. (H,H)
Security	Access control	A member accesses notifications, and the system ensures only authorized users can view their notifications. (H,H)
Maintainability	Notification feature updates	Developers modify notification logic without affecting other system components. (M,M)
Flexibility	Notification extensibility	The system supports adding new notification types without major redesign. (M,M)
Supportability	Event logging	Notification events and delivery failures are logged to support debugging and monitoring. (M,M)
Testability	Notification verification	Notification delivery and display can be validated through automated and manual tests. (M,M)

2.10 Utility Tree for Member Private Messaging

Table 17: Utility Tree for Private Messaging

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Messaging ease	A member sends and receives private messages, and the system allows communication through a simple and intuitive interface. (H,M)
	Conversation readability	A member views a conversation, and messages are displayed in a clear and chronological order. (H,M)
Performance	Message delivery latency	A member sends a message under normal load, and the system delivers it to the recipient within 1 second. (H,H)
	Conversation load time	A member opens a conversation, and messages are loaded within 1 second. (H,M)
Security	Message privacy	A member sends a private message, and the system ensures that only the intended recipient can access the message content. (H,H)
	Access control	A member attempts to access a conversation, and only authorized participants can view or interact with it. (H,H)
Reliability	Message delivery guarantee	A member sends a message, and the system ensures the message is delivered without loss or duplication. (H,H)
	Message consistency	A member views messages, and the system consistently displays the correct conversation history. (H,H)
Availability	Messaging service availability	A member sends or receives messages at any time, and the messaging system is available with 99.9% uptime. (H,H)
Scalability	Concurrent messaging	At peak load, many members send messages simultaneously, and the system maintains stable performance. (H,H)
Maintainability	Messaging feature updates	Developers modify messaging functionality without affecting other system components. (M,M)
Flexibility	Messaging extensibility	The system supports adding new messaging features (e.g., attachments) without major redesign. (M,M)
Supportability	Message logging	Messaging operations and failures are logged to support debugging and monitoring. (M,M)
Testability	Messaging verification	Message sending, receiving, and display can be validated through automated and manual tests. (M,M)

2.11 Utility Tree for Member Reporting Inappropriate Content

Table 18: Utility Tree for Reporting System

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Reporting ease	A member reports inappropriate content, and the system allows submission in a few simple steps with clear feedback. (H,M)
	Feedback clarity	A member submits a report, and the system confirms successful submission with clear status information. (H,M)
Performance	Report submission time	A member submits a report under normal load, and the system processes it within 1 second. (H,M)
	Report retrieval time	A moderator accesses reports, and the system loads them within 2 seconds. (M,M)
Security	Access control	A member attempts to submit a report, and only authenticated users are allowed to perform the action. (H,H)
	Abuse prevention	A member submits reports, and the system prevents spam or excessive reporting through validation mechanisms. (H,H)
Reliability	Report persistence	A member submits a report, and the system ensures the report is stored correctly without loss. (H,H)
	Processing consistency	Reports are processed by moderators, and the system maintains consistent status updates without errors. (H,M)
Availability	Reporting service availability	A member submits reports at any time, and the reporting system is available with 99.9% uptime. (H,H)
Scalability	High report volume	At peak load, many members submit reports simultaneously, and the system maintains stable performance. (M,M)
Maintainability	Reporting feature updates	Developers modify reporting functionality without affecting other system components. (M,M)
Flexibility	Report extensibility	The system supports adding new report categories or workflows without major redesign. (M,M)
Supportability	Report logging	Reporting actions and errors are logged to support debugging and moderation analysis. (M,M)
Testability	Reporting verification	Report submission and processing can be validated through automated and manual tests. (M,M)

3 For Community Owners and Moderators

3.1 Utility Tree for Community Creation

Table 19: Utility Tree for Community Creation

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Creation ease	A community owner creates a new community, and the system allows completion in a few intuitive steps with clear guidance. (H,M)
	Input feedback	A community owner submits community details, and the system provides immediate and clear validation feedback for missing or invalid inputs. (H,M)
Performance	Creation response time	A community owner creates a community under normal load, and the system processes and confirms creation within 2 seconds. (H,M)
	Page load time	A community owner accesses the community creation interface, and the page loads within 2 seconds. (M,M)
Security	Access control	A user attempts to create a community, and only authorized members are allowed to perform the action. (H,H)
	Data validation	A community owner submits community information, and the system validates inputs to prevent malicious or inappropriate data. (H,H)
Reliability	Creation consistency	A community owner creates a community, and the system ensures the community is stored and initialized correctly without errors. (H,H)
Availability	Creation service availability	A community owner attempts to create a community at any time, and the system is available with 99.9% uptime. (H,H)
Scalability	Concurrent creation	At peak load, multiple users create communities simultaneously, and the system maintains stable performance. (M,M)
Maintainability	Community feature updates	Developers modify community creation features without affecting other system components. (M,M)
Flexibility	Community configuration extensibility	The system supports adding new community settings without major redesign. (M,M)
Supportability	Creation logging	Community creation actions and failures are logged for debugging and monitoring. (M,M)
Testability	Creation verification	Community creation functionality can be validated through automated and manual tests. (M,M)

3.2 Utility Tree for Defining Community Rules

Table 20: Utility Tree for Community Rules Management

Quality Attribute	Attribute Refinement	ASR
Usability	Rule creation ease	A community owner defines or updates community rules, and the system allows completion in a few intuitive steps with clear guidance. (H,M)
	Rule clarity	A member views community rules, and the system displays them in a clear and readable format. (H,M)
Performance	Rule update response	A community owner updates rules under normal load, and the system processes and saves changes within 1 second. (H,M)
	Rule retrieval time	A member accesses community rules, and the system loads them within 1 second. (H,M)
Security	Access control	A user attempts to create or modify rules, and only the community owner or authorized moderators can perform the action. (H,H)
	Content validation	A community owner submits rule content, and the system validates and sanitizes inputs to prevent malicious or harmful content. (H,H)
Reliability	Rule consistency	A community owner updates rules, and the system ensures all changes are stored and displayed correctly without data loss. (H,H)
Availability	Rule service availability	Users access community rules at any time, and the system is available with 99.9% uptime. (H,H)
Scalability	Concurrent access	At peak load, many users view community rules simultaneously, and the system maintains stable performance. (M,M)
Maintainability	Rule management updates	Developers modify rule management features without affecting other system components. (M,M)
Flexibility	Rule extensibility	The system supports adding new rule formats or categories without major redesign. (M,M)
Supportability	Rule change logging	Rule updates and related errors are logged to support debugging and moderation tracking. (M,M)
Testability	Rule verification	Rule creation, modification, and display can be validated through automated and manual tests. (M,M)

3.3 Utility Tree for Community Privacy Settings

Table 21: Utility Tree for Community Privacy Control

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Setting configuration ease	A community owner sets or updates privacy settings (public or private), and the system allows completion in a few intuitive steps with clear feedback. (H,M)
	Privacy visibility clarity	A user views a community, and the system clearly indicates whether it is public or private. (H,M)
Performance	Setting update response	A community owner changes privacy settings under normal load, and the system applies changes within 1 second. (H,M)
	Access validation time	A user attempts to access a community, and the system validates access permissions within 0.5 seconds. (H,M)
Security	Access enforcement	A user attempts to access a private community, and the system restricts access to authorized members only. (H,H)
	Unauthorized access prevention	A non-member attempts to view restricted content, and the system prevents exposure of private data. (H,H)
Reliability	Policy consistency	A community owner updates privacy settings, and the system consistently enforces the updated policy across all access points. (H,H)
Availability	Privacy control availability	A community owner updates or users access community privacy settings at any time, and the system is available with 99.9% uptime. (H,H)
Scalability	Access control under load	At peak load, many users attempt to access communities simultaneously, and the system enforces privacy rules without performance degradation. (H,M)
Maintainability	Privacy feature updates	Developers modify privacy control logic without affecting other system components. (M,M)
Flexibility	Privacy model extensibility	The system supports future extensions (e.g., invite-only or role-based access) without major redesign. (M,M)
Supportability	Access logging	Access attempts and violations are logged to support monitoring and debugging. (M,M)
Testability	Access rule verification	Privacy enforcement can be validated through automated and manual test cases. (M,M)

3.4 Utility Tree for Assigning Moderator Roles

Table 22: Utility Tree for Assigning Moderator Roles to Members

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Role assignment simplicity	A community owner assigns or removes moderator roles, and the system allows completion through a clear and intuitive interface within a few steps. (H,M)
	Role visibility clarity	A user views member roles in a community, and the system clearly displays who is a moderator or owner. (H,M)
Performance	Role assignment response time	A community owner assigns a moderator under normal load, and the system updates the role within 1 second. (H,M)
	Permission propagation time	A member is assigned a moderator role, and the system applies new permissions across all relevant features within 1 second. (H,M)
Security	Role authorization enforcement	A non-owner attempts to assign moderator roles, and the system denies the action and logs the attempt. (H,H)
	Privilege misuse prevention	A moderator attempts to perform actions beyond their permissions, and the system restricts unauthorized operations. (H,H)
Reliability	Role consistency	A moderator role is assigned or revoked, and the system consistently reflects the correct role across all services without conflict. (H,H)
Availability	Role management availability	Community owners manage moderator roles at any time, and the system remains available with 99.9% uptime. (H,H)
Scalability	Role assignment under load	At peak usage, multiple communities assign moderators simultaneously, and the system maintains stable performance. (H,M)
Maintainability	Role management extensibility	Developers update role management logic without impacting other modules such as permissions or user profiles. (M,M)
Flexibility	Role model extensibility	The system supports adding new roles (e.g., senior moderator) without major redesign. (M,M)
Supportability	Role change logging	All role assignment and revocation actions are logged for auditing and debugging purposes. (M,M)
Testability	Role assignment verification	Role assignment and permission enforcement can be validated through automated and manual test cases. (M,M)

3.5 Utility Tree for Reviewing and Approving Posts

Table 23: Utility Tree for Moderation of Pending Posts

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Moderation workflow clarity	A moderator reviews pending posts, and the system provides a clear interface to approve or reject content with minimal steps. (H,M)
	Content review readability	A moderator views a pending post, and the system displays content (text, images, metadata) clearly for quick decision-making. (H,M)
Performance	Review action response time	A moderator approves or rejects a post under normal load, and the system processes the action within 1 second. (H,M)
	Queue loading time	A moderator opens the moderation queue, and the system loads pending posts within 2 seconds. (H,M)
Security	Moderation authorization enforcement	A non-moderator attempts to approve or reject posts, and the system denies access and logs the attempt. (H,H)
	Content integrity protection	A post under review cannot be altered by unauthorized users before moderation is completed. (H,H)
Reliability	Moderation decision consistency	A post is approved or rejected, and the system consistently updates its status across all views without duplication or loss. (H,H)
Availability	Moderation system availability	Moderators access and process pending posts at any time, and the system maintains 99.9% uptime. (H,H)
Scalability	Moderation queue under load	During peak activity, a large number of posts enter the moderation queue, and the system handles the load without degrading performance. (H,M)
Maintainability	Moderation logic maintainability	Developers modify moderation workflows without affecting unrelated components such as posting or notifications. (M,M)
Flexibility	Moderation rule extensibility	The system supports adding automated moderation rules (e.g., AI filtering) without major redesign. (M,M)
Supportability	Moderation activity logging	All moderation actions (approve/reject) are logged for auditing and debugging. (M,M)
Testability	Moderation workflow testing	Moderation scenarios (approve, reject, edge cases) can be validated through automated and manual test cases. (M,M)

3.6 Utility Tree for Removing Posts or Comments

Table 24: Utility Tree for Removing Posts or Comments

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Removal action simplicity	A moderator removes a post or comment, and the system allows completion in a few clear steps with confirmation feedback. (H,M)
	Removal feedback clarity	A user attempts to view removed content, and the system clearly indicates that it has been removed due to policy violations. (H,M)
Performance	Removal response time	A moderator removes content under normal load, and the system processes the action within 1 second. (H,M)
	Content update propagation	Removed content is reflected across feeds and views within 1 second. (H,M)
Security	Removal authorization enforcement	A non-moderator attempts to remove content, and the system denies the action and logs the attempt. (H,H)
	Unauthorized restoration prevention	A removed post or comment cannot be restored by unauthorized users. (H,H)
Reliability	Removal consistency	Once content is removed, the system consistently prevents it from appearing across all interfaces. (H,H)
Availability	Moderation availability	Moderators can remove content at any time with 99.9% system uptime. (H,H)
Scalability	Bulk moderation under load	During peak usage, moderators remove large volumes of content, and the system maintains stable performance. (H,M)
Maintainability	Content moderation maintainability	Developers update removal logic without impacting other modules. (M,M)
Flexibility	Moderation policy flexibility	The system supports evolving content policies without major redesign. (M,M)
Supportability	Removal logging	All removal actions are logged for auditing and debugging. (M,M)
Testability	Removal testing	Content removal scenarios can be verified through automated and manual tests. (M,M)

3.7 Utility Tree for Muting Users from Community

Table 25: Utility Tree for Muting Users

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Mute action simplicity	A moderator mutes a user, and the system provides an intuitive interface to set duration and confirm the action. (H,M)
	Mute status visibility	A muted user attempts to interact, and the system clearly informs them of the restriction and duration. (H,M)
Performance	Mute action response	A moderator mutes a user, and the system enforces the restriction within 1 second. (H,M)
	Restriction enforcement latency	A muted user attempts to post or comment, and the system blocks the action within 0.5 seconds. (H,M)
Security	Mute authorization enforcement	A non-moderator attempts to mute users, and the system denies the action. (H,H)
	Restriction bypass prevention	A muted user attempts to bypass restrictions, and the system prevents unauthorized interactions. (H,H)
Reliability	Mute duration accuracy	A user is muted for a specified duration, and the system enforces and lifts the restriction accurately. (H,H)
Availability	Mute functionality availability	Moderators can mute users at any time with 99.9% uptime. (H,H)
Scalability	User restriction under load	The system enforces mute restrictions for many users simultaneously without degradation. (H,M)
Maintainability	Restriction logic maintainability	Developers update mute logic without affecting other moderation features. (M,M)
Flexibility	Restriction policy extensibility	The system supports additional restriction types (e.g., partial mute) without major redesign. (M,M)
Supportability	Mute action logging	All mute actions and durations are logged for monitoring and debugging. (M,M)
Testability	Mute enforcement testing	Mute scenarios (apply, expire, edge cases) can be validated through tests. (M,M)

3.8 Utility Tree for Banning Users from Community

Table 26: Utility Tree for Banning Users

Quality Attribute	Attribute Refinement	ASR
Usability	Ban action clarity	A moderator bans a user, and the system provides clear steps including reason input and confirmation. (H,M)
	Ban notification clarity	A banned user attempts to access the community, and the system clearly communicates the ban status and reason. (H,M)
Performance	Ban enforcement time	A moderator bans a user, and the system enforces the restriction across all features within 1 second. (H,M)
	Access denial response	A banned user attempts to access content, and the system denies access within 0.5 seconds. (H,M)
Security	Ban authorization enforcement	A non-moderator attempts to ban users, and the system denies the action and logs it. (H,H)
	Re-entry prevention	A banned user attempts to rejoin or bypass the ban, and the system prevents unauthorized access. (H,H)
Reliability	Ban consistency	A banned user is consistently restricted from all community interactions without exception. (H,H)
Availability	Ban management availability	Moderators manage bans at any time with 99.9% uptime. (H,H)
Scalability	Ban enforcement at scale	The system enforces bans for a large number of users across communities without performance degradation. (H,M)
Maintainability	Ban logic maintainability	Developers update banning mechanisms without impacting other system modules. (M,M)
Flexibility	Ban policy extensibility	The system supports temporary or conditional bans without major redesign. (M,M)
Supportability	Ban audit logging	All ban actions, reasons, and timestamps are logged for auditing purposes. (M,M)
Testability	Ban enforcement testing	Ban scenarios (apply, revoke, bypass attempts) can be validated through testing. (M,M)

4 For System Administrators

4.1 Utility Tree for System Analytics Dashboard

Table 27: Utility Tree for System Analytics Dashboard

Quality Attribute	Attribute Refinement	ASR
Usability	Dashboard readability	A system admin views the analytics dashboard, and the system presents metrics in a clear and organized manner. (H,M)
	Navigation efficiency	A system admin navigates between analytics sections, and the system allows quick and intuitive access. (H,M)
Performance	Dashboard load time	A system admin opens the dashboard, and the system loads key metrics within 2 seconds. (H,M)
	Data refresh latency	Analytics data updates in near real-time with a delay of no more than 5 seconds. (H,M)
Security	Access control	A non-admin attempts to access the dashboard, and the system denies access. (H,H)
Reliability	Data accuracy	The dashboard displays accurate and consistent analytics data across sessions. (H,H)
Availability	Dashboard availability	The analytics dashboard is accessible with 99.9% uptime. (H,H)
Scalability	Analytics under load	The system handles large volumes of analytics data without performance degradation. (H,M)
Maintainability	Analytics maintainability	Developers update dashboard components without affecting other system modules. (M,M)
Flexibility	Metric extensibility	New analytics metrics can be added without major redesign. (M,M)
Supportability	Monitoring logs	System metrics and access logs are recorded for debugging and monitoring. (M,M)
Testability	Analytics validation	Dashboard metrics and data pipelines can be verified through tests. (M,M)

4.2 Utility Tree for Reviewing Escalated Reports

Table 28: Utility Tree for Reviewing Escalated Reports

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Report review clarity	A system admin reviews escalated reports, and the system presents all relevant information clearly. (H,M)
	Decision workflow simplicity	A system admin takes action (dismiss, escalate, ban), and the system provides a clear workflow. (H,M)
Performance	Report loading time	A system admin opens a report queue, and the system loads within 2 seconds. (H,M)
	Action processing time	Actions on reports are processed within 1 second. (H,M)
Security	Report access control	Unauthorized users cannot access escalated reports. (H,H)
Reliability	Report consistency	Report status updates are consistent across all views. (H,H)
Availability	Report system availability	The reporting system is available with 99.9% uptime. (H,H)
Scalability	Report handling under load	The system processes large volumes of reports without degradation. (H,M)
Maintainability	Reporting system maintainability	Developers update reporting workflows without affecting other modules. (M,M)
Flexibility	Action extensibility	The system supports adding new moderation actions without redesign. (M,M)
Supportability	Report audit logs	All report actions are logged for auditing. (M,M)
Testability	Report workflow testing	Report handling scenarios can be validated through tests. (M,M)

4.3 Utility Tree for Global User Banning

Table 29: Utility Tree for Global User Banning

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Ban workflow clarity	A system admin bans a user globally, and the system provides clear steps and confirmation. (H,M)
	Ban communication clarity	A banned user is clearly informed of the global restriction. (H,M)
Performance	Ban enforcement time	A global ban is enforced across all communities within 1 second. (H,M)
	Access denial latency	A banned user is denied access within 0.5 seconds. (H,M)
Security	Admin authorization enforcement	Only authorized admins can perform global bans. (H,H)
Reliability	Ban consistency	A globally banned user is restricted across all services without exception. (H,H)
Availability	Ban system availability	Global banning functionality is available with 99.9% uptime. (H,H)
Scalability	Global enforcement at scale	The system enforces bans across millions of users without degradation. (H,M)
Maintainability	Ban system maintainability	Developers update ban logic without affecting other modules. (M,M)
Flexibility	Ban policy extensibility	The system supports temporary or conditional global bans. (M,M)
Supportability	Ban audit logs	All global bans are logged with reasons and timestamps. (M,M)
Testability	Ban system testing	Global ban scenarios can be validated through testing. (M,M)

4.4 Utility Tree for Removing or Restricting Communities

Table 30: Utility Tree for Community Removal or Restriction

Quality Attribute	**Attribute Refinement**	**ASR**
Usability	Action clarity	A system admin removes or restricts a community, and the system provides clear steps and confirmation. (H,M)
	Status visibility	Users viewing a restricted community see clear status indicators. (H,M)
Performance	Action processing time	Community restriction/removal is processed within 1 second. (H,M)
	Propagation time	Changes are reflected across the system within 1 second. (H,M)
Security	Admin authorization	Only authorized admins can remove or restrict communities. (H,H)
Reliability	Enforcement consistency	Restricted communities are consistently controlled across all features. (H,H)
Availability	Control system availability	Community control features are available with 99.9% uptime. (H,H)
Scalability	Enforcement under load	The system handles large numbers of community restrictions without degradation. (H,M)
Maintainability	Control logic maintainability	Developers update restriction logic without affecting unrelated modules. (M,M)
Flexibility	Policy extensibility	The system supports different restriction levels without redesign. (M,M)
Supportability	Action logging	All admin actions on communities are logged for auditing. (M,M)
Testability	Control testing	Community restriction/removal scenarios can be validated through testing. (M,M)